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Abstract— 2D 3D NeuroLocMap—— - 3D 4 - - IMU FCC 3D 10 EKF Town05/03/07/10HD KITTI 07 10 EKF —— Sim(3) 16.5–59.3% SE(3) 1.5–33.9%—— 33.7–68.8% EKF 9 44.24 m 398.05 m Sim(3) CPU 31 FPS

Index Terms— - -

I.

3D [1] SLAM IMU GNSS [2] SLAM 3D [3], [4], [5], [6], [7] SLAM HART - 185 - 2D 3D EKF NeuroLocMap—— 3D 4 $x\ y\ z\ \psi$ - IMU 3D ψ $< 5^\circ$ 4 - [8], [9]

- - 4
- 5 RatSLAM 929 185
- - CPU 31 FPS ——
——
- CARLA Town05 / 03 / 07 / 10HD KITTI 07 Sim(3) +16.5–59.3% SE(3) +1.5–33.9% VO +33.7–68.8% EKF 9 44.24 m 398.05 m Sim(3)

II.

-

A. SLAM

SLAM Mur-Artal ORB-SLAM [10] ORB Engel LSD-SLAM [11] DSO [12] Newman FAB-MAP [13] Arandjelovic NetVLAD [14] SLAM >100 DSO — —ORB-SLAM DSO IMU SLAM Mourikis Roumeliotis MSCKF [15] - Qin VINS-Mono [16] Mahony [17] Madgwick [18] —— IMU VINS-Mono IMU [19], [20]

B.

O'Keefe Dostrovsky [21] Hafting [22] Taube [23] Moser [24] Finkelstein 3D FCC 3D [6] Milford RatSLAM [8], [26] 66 km Steckel Peremans BatSLAM [27] SLAM Yu NeuroSLAM CNN 3D Lowry [28] - (1) 2D 3D / 4 (2) RatSLAM 1D 5 (3) IMU - FCC 3D 4 185

C.

Arandjelovic NetVLAD [14] VLAD Goodale Milner [29] Mishkin [30] CABLA V1→V4→IT [32] Vaswani Transformer [33] NetVLAD Google Street View Transformer $\mathcal{O}(N^2)$ / — - 5 RatSLAM 929 185 31 FPS

D.

I SLAM (1) 2D RatSLAM BatSLAM 3D RatSLAM 1-D (3) IMU (4) [2] SLAM NeuroLocMap 3D FCC 4 HART - 185 - EKF

III.

A.

. ^{vis} ^{imu} f g h t . s VO 10 EKF $\log s$ s^* $\hat{z}_t$ DAv2 g_θ III-E $\boldsymbol{e}_t$ III-F

B.

10 EKF - - - EKF

1) : t

1)

$$\mathbf{U}_t = \left[I_t,\, \mathbf{a}_t^\top,\, \boldsymbol{\omega}_t^\top \right] \tag{1}$$

I_t RGB $\mathbf{a}_t$ $\boldsymbol{\omega}_t$

2) 4

$$\mathbf{X}_t = [x_t,\, y_t,\, z_t,\, \psi_t]^\top \in \mathbb{R}^3 \times \mathrm{SO}(2) \tag{2}$$

$(x_t, y_t, z_t) \in \mathbb{R}^3$ 3D $\psi_t \in [-\pi, \pi)$

3) Cummins

$$\mathbf{X}_{t+1} = \mathcal{F}\left(\mathbf{X}_t,\, \mathbf{U}_t\right) \tag{3}$$

$\mathcal{F}(\cdot)$ $\mathcal{F}(\cdot)$ 3.3 - 3.4

1

1. /
2. IMU
3. IIR
4. -
5. MSCKF III-E EKF - III-F ——

VO up-to-scale VIO

C.

RGB . RGB $I_t \in \mathbb{R}^{H \times W \times 3}$ 1 LGN . LGN

$$Y_t = \mathcal{G}(I_t) = 0.299R + 0.587G + 0.114B \tag{4}$$

$$U_t = \mathcal{C}_\lambda(Y_t), \quad \lambda = 0.02 \tag{5}$$

$$V_t = U_t * \mathcal{K}_\sigma, \quad \sigma = 0.5 \tag{6}$$

$\mathcal{G}$ - $\mathcal{C}_\lambda$ CLAHE λ $\mathcal{K}_\sigma$ σ CLAHE

TABLE I

RatSLAM [26]	2D	PC+HDC		-
BatSLAM [27]	2D	PC+HDC		-
NeuroSLAM	3D	3D-GC	CNN	LSTM
NeuroLocMap	4D	3D-GC+M-HDC	+HART	Trans.

: PC: ; HDC: ; GC: ; M-HDC: ; : ; : ; : ; Trans.: Transformer ; : ; +HART: HART / : $\mathcal{O}(N)$ Query-Key-Value $\mathcal{O}(N^2)$

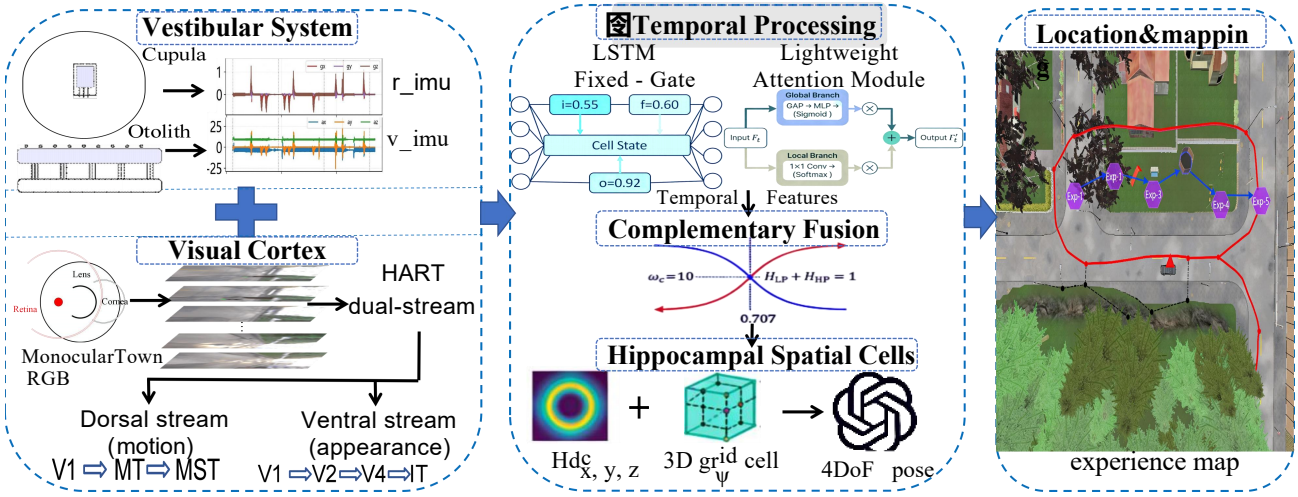


Fig. 1. - RGB IMU

2 V1-V4 .

$$\tilde{V}_t = \frac{V_t - \min(V_t)}{\max(V_t) - \min(V_t)} \tag{7}$$

$$Z_t = \mathcal{D}(\tilde{V}_t) \in \mathbb{R}^{64\times 64} \tag{8}$$

$$Z_t(i,j) \leftarrow Z_t(i,j) - \frac{1}{64}\sum_{k=1}^{64} Z_t(i,k) \tag{9}$$

$\mathcal{D}$ V1 V2

3 IT .

$$\boldsymbol{\phi}_t = \mathrm{vec}(Z_t) \in \mathbb{R}^{4096} \tag{10}$$

$\mathrm{vec}(\cdot)$ $\boldsymbol{\phi}_t$ t $64 \times 64 = 4096$ IT

. SAD

$$d_{\cos}(\boldsymbol{\phi}_t, \boldsymbol{\phi}_i) = 1 - \frac{\boldsymbol{\phi}_t^\top \boldsymbol{\phi}_i}{\|\boldsymbol{\phi}_t\|\,\|\boldsymbol{\phi}_i\|} \tag{11}$$

VT

$$\min_{i\in\mathcal{VT}} d_{\cos}(\boldsymbol{\phi}_t, \boldsymbol{\phi}_i) > \tau_{vt} \quad \text{VT}, \quad \tau_{vt} = 0.08 \tag{12}$$

$\tau_{vt} = 0.08$ OpenRatSLAM 0.15 . (1) / (2)

(2) 4 $\mathbf{X}_t$

. [29], [30] V1 "where" V1→MT→MST

$$\mathbf{F}_t^{\text{dorsal}} = 0.6 \cdot \|\nabla Z_t\| + 0.4 \cdot Z_t \tag{13}$$

$\|\nabla Z_t\|$ Sobel MT/MST "what" V1→V4→IT

$$\mathbf{F}_t^{\text{ventral}} = 0.4 \cdot Z_t + 0.3 \cdot \sigma_{\text{local}}(Z_t) + 0.3 \cdot \mathcal{C}_\lambda(Z_t) \tag{14}$$

σ_{local} $\mathcal{C}_\lambda$ V4 IT - II-C Query-Key-Value $\mathcal{O}(N)$ LSTM. LSTM - [19], [20]

$$: i_t = 0.55 \tag{15}$$

$$: f_t = 0.60 \quad 60\% \tag{16}$$

$$: o_t = 0.92 \tag{17}$$

i_t f_t o_t LSTM SLAM (1) SLAM (2) $f_t = 0.6$ $o_t =$ $\tau_{vt} = 0.08$ 39 0.92 195 $i_t = 0.55$ LSTM 31 FPS FPGA

LSTM - $\mathbf{H}_t$ $\mathbf{C}_t$

$$\mathbf{C}_t = f_t \odot \mathbf{C}_{t-1} + i_t \odot \tanh(\mathbf{F}_t^{\text{fused}}) \quad (18)$$

$$\mathbf{H}_t = o_t \odot \tanh(\mathbf{C}_t) \quad (19)$$

$\odot$ Hadamard $\mathbf{C}_t$ LSTM $\mathbf{H}_t$ $\mathbf{F}_t^{\text{fused}}$
. IV LSTM NeuroSLAM LSTM RMSE 55% .

—

. MST
. $\mathbf{K} \in \mathbb{R}^{3\times3}$ N $\mathbf{p}_i^t \in \mathbb{R}^2$ KLT

$$\tilde{\mathbf{q}}_i^t = \mathbf{K}^{-1}\begin{bmatrix}\mathbf{p}_i^t \\ 1\end{bmatrix}, \qquad \mathbf{q}_i^t = \pi(\tilde{\mathbf{q}}_i^t) = \frac{1}{\tilde{q}_{i,3}^t}\begin{bmatrix}\tilde{q}_{i,1}^t \\ \tilde{q}_{i,2}^t\end{bmatrix} \quad (20)$$

π

$$\mathbf{f}_i = \mathbf{q}_i^t - \mathbf{q}_i^{t-1}, \quad \mathbf{q}_i \equiv \mathbf{q}_i^{t-1} \quad (21)$$

. $\mathbf{q} = (x, y)$ $\Delta t\, r\, [-y,\, x]^\top$

$$r_t^{\text{vis}} = \frac{\sum_{i=1}^N w_i (x_i f_{i,y} - y_i f_{i,x})}{\Delta t \sum_{i=1}^N w_i (x_i^2 + y_i^2)} \quad (22)$$

w_i Huber N r_t^{vis}
.

$$\mathbf{f}_i^\perp = \mathbf{f}_i - \Delta t\, r_t^{\text{vis}} \begin{bmatrix}-y_i \\ x_i\end{bmatrix} \quad (23)$$

$\lambda_i \approx 1/Z_g$

$$\mathbf{f}_i^\perp \approx \Delta t\, \lambda_i \underbrace{\begin{bmatrix}-1 & 0 & x_i \\ 0 & -1 & y_i\end{bmatrix}}_{\mathbf{A}_i} \mathbf{v}_t^{\text{vis}}, \quad \mathbf{v}_t^{\text{vis}} = (v_x^{\text{vis}}, v_y^{\text{vis}}, v_z^{\text{vis}})^\top \quad (24)$$

w_i

$$\bar{\mathbf{v}}_t^{\text{vis}} = \Big(\sum_i w_i\, (\Delta t\, \lambda_i)^2\, \mathbf{A}_i^\top \mathbf{A}_i\Big)^{-1} \sum_i w_i\, (\Delta t\, \lambda_i)\, \mathbf{A}_i^\top \mathbf{f}_i^\perp \quad (25)$$

κ_t

$$\mathbf{v}_t^{\text{vis}} = \kappa_t\, \bar{\mathbf{v}}_t^{\text{vis}} \quad (26)$$

λ_i i $\mathbf{A}_i$ $\bar{\mathbf{v}}_t^{\text{vis}}$ κ_t
$\hat{z}_t$ DAv2 $\boldsymbol{e}_t$ III-E
.
. $\mathbf{g}_t \in \mathbb{R}^3$ IMU z / 4 -

$$r_t \approx \mathbf{e}_z^\top\, \boldsymbol{\omega}_t^g, \quad \mathbf{e}_z = (0,0,1)^\top. \quad (27)$$

$\boldsymbol{\omega}_t^g$ IMU $\mathbf{e}_z$
. z g=9.81 Δt

$$v_{trans}^{\text{imu}}(t) \approx \max\big(0, (\|\mathbf{a}_t\| - g)\, \Delta t\, k_t\big), \quad (28)$$

$$v_z^{\text{imu}}(t) \approx (a_{t,z} - g)\, \Delta t\, k_z, \quad (29)$$

$g = 9.81\,\text{m/s}^2$ k_t, k_z Δt /
. /
IMU- . IMU
. -
. $\mathcal{L}_\tau$ τ $\mathcal{H}_\tau$ $\mathcal{L}_\tau + \mathcal{H}_\tau = \text{Id}$

$$: \; L_t = \alpha L_{t-1} + (1-\alpha) x_t, \quad L_0 = 0 \quad (30)$$

$$: \; H_t = x_t - L_t \quad (31)$$

α IIR L_t H_t
. 7a —

—

η η_r
. Δt IIR $\alpha = \exp(-\Delta t/\tau)$ $\alpha_r = \exp(-\Delta t/\tau_r)$ x_t

$$: \; L_t = \alpha L_{t-1} + (1-\alpha) x_t, \quad L_0 = 0 \quad (32)$$

$$: \; H_t = x_t - L_t \quad (33)$$

$$\mathbf{v}_t^{\text{f}} = (1-\eta)\, L_t^{\text{vis}} + \eta\, H_t^{\text{imu}} \quad (34)$$

$$r_t^{\text{f}} = (1-\eta_r)\, L_t^{r,vis} + \eta_r\, H_t^r \quad (35)$$

L_t^{vis} H_t^{imu} IMU
. $s = j\omega$ —

$$H_{LP}(s) = \frac{1}{1+\tau s} = \frac{1/\tau}{s+1/\tau}, \quad (36)$$

$$H_{HP}(s) = \frac{\tau s}{1+\tau s} = \frac{s}{s+1/\tau}, \quad (37)$$

$H_{LP}(s) + H_{HP}(s) = 1$ $\omega_c = 1/\tau$ $\omega \ll \omega_c$ $|H_{LP}| \approx 1$ $|H_{HP}| \approx 0$ $\omega \gg \omega_c$ $|H_{LP}| \approx 0$ $|H_{HP}| \approx 1$ IMU
$\tau = 0.1$ s $\omega_c = 10$ rad/s ≈ 1.6 Hz IMU >2 Hz <1 Hz - 0.1-10 Hz 0.5 Hz [19]
. L_t L_{t-1} $\mathcal{O}(1)$ — — - $\mathcal{O}(n^3)$
. η η_r IMU $\eta \to 1$ IMU $\eta \to 0$

$$\eta_t = f(\sigma_t^{\text{vis}}, \sigma_t^{\text{imu}}), \qquad \eta_t = \frac{\sigma_t^{\text{vis}}}{\sigma_t^{\text{vis}} + \sigma_t^{\text{imu}}} \quad (38)$$

σ_t^{vis} σ_t^{imu} η_t
(26) κ_t

$$\kappa_{t+1} = \kappa_t \cdot \left(1 + \beta \frac{\|\mathbf{v}_t^{\text{imu}}\| - \|\kappa_t \bar{\mathbf{v}}_t^{\text{vis}}\|}{\|\kappa_t \bar{\mathbf{v}}_t^{\text{vis}}\|}\right) \quad (39)$$

$\beta \ll 1$ $\beta = 0.01$

D.

conjunctive pose cells 4 3D (x, y, z) 3D MD-CAN ψ 3D . 3D 3D 3D FCC 3D [6] [4] 2D 3D . 3D $N_x \times N_y \times N_z$ $N_x = N_y = N_z = 61$ 3D MD-CAN $\mathbf{A}_t^{\text{g}} \in \mathbb{R}^{N_x \times N_y \times N_z}$ 3D (x_t, y_t, z_t) (i, j, k) “ ” FCC ——3D —— FCC 12 3 (a) (x, y, z) 3D (b) FCC 2D 4
. “ ”
(1) .

$$\mathcal{K}_{a,b,c}^{\text{g}} = \mathcal{N}(a; 0, \sigma_x)\mathcal{N}(b; 0, \sigma_y)\mathcal{N}(c; 0, \sigma_z) \quad (40)$$

$\mathcal{N}(\cdot; \mu, \sigma)$ $\sigma_x, \sigma_y, \sigma_z$

$$\Delta A_{i,j,k}^{\text{g}} = \sum_{p,q,r} A_{p,q,r}^{\text{g}}\, \mathcal{K}_{\langle i-p\rangle_{N_x}, \langle j-q\rangle_{N_y}, \langle k-r\rangle_{N_z}}^{\text{g}} \quad (41)$$

$\langle\cdot\rangle_N$ N $\sigma_x = \sigma_y = \sigma_z = 5$
(2) · 2

$$A_{i,j,k}^{\text{g}} \leftarrow \big[A_{i,j,k}^{\text{g}} + \Delta A_{i,j,k}^{\text{g}} - \lambda\, \bar{A}^{\text{g}}\big]_+, \quad \bar{A}^{\text{g}} = \frac{1}{N_x N_y N_z} \sum_{p,q,r} A_{p,q,r}^{\text{g}} \quad (42)$$

$[\cdot]_+ = \max(0, \cdot)$ λ $\lambda = 1.0$ L_1

$$A^{\text{g}} \leftarrow \frac{A^{\text{g}}}{\|A^{\text{g}}\|_1} = \frac{A^{\text{g}}}{\sum_{i,j,k} |A_{i,j,k}^{\text{g}}|} \quad (43)$$

(3) . $\mathbf{v}_t^{\text{f}} = (v_x^f, v_y^f, v_z^f)$

$$(\Delta i, \Delta j, \Delta k) = \big(\lfloor \gamma_x v_x^f \Delta t \rfloor,\ \lfloor \gamma_y v_y^f \Delta t \rfloor,\ \lfloor \gamma_z v_z^f \Delta t \rfloor\big) \quad (44)$$

$\gamma_x, \gamma_y, \gamma_z$ -

$$A_{i,j,k}^{\text{g}} \leftarrow A_{\langle i-\Delta i\rangle_{N_x}, \langle j-\Delta j\rangle_{N_y}, \langle k-\Delta k\rangle_{N_z}}^{\text{g}} \quad (45)$$

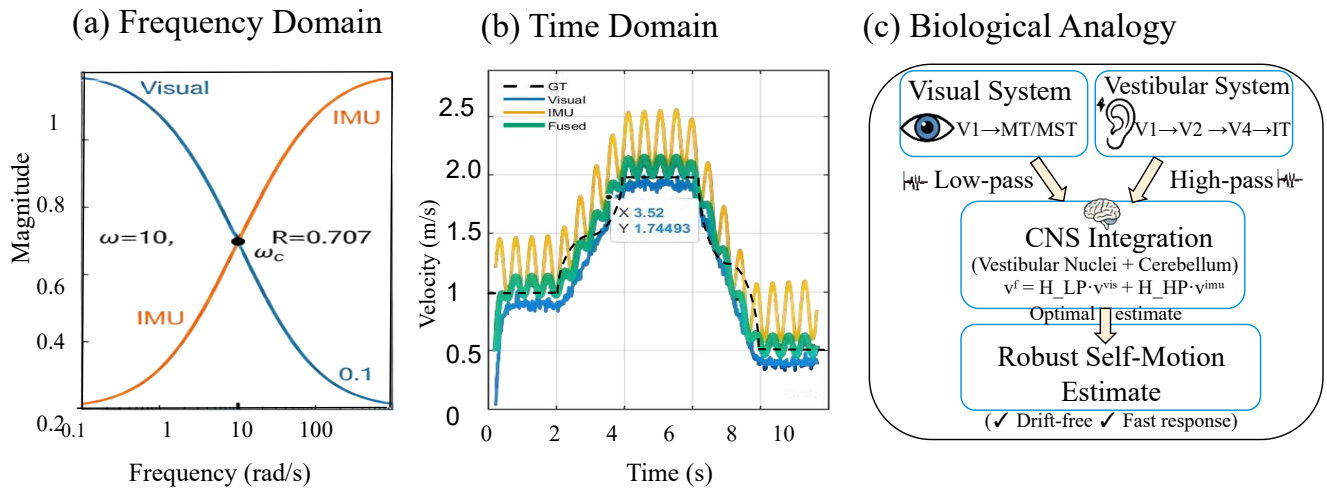


Fig. 2. IMU- (a) (b) (c) -

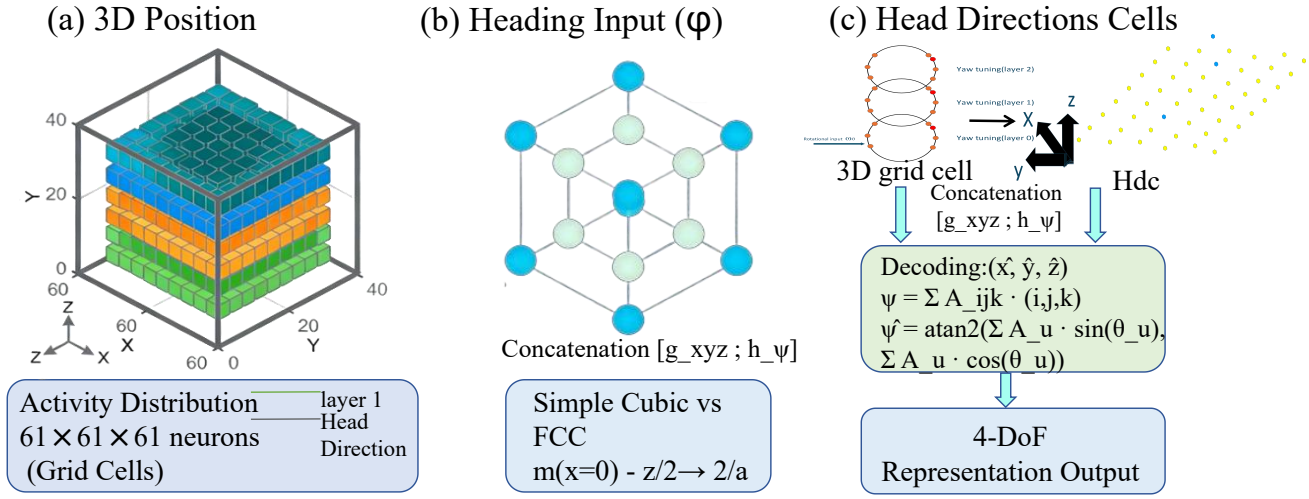


Fig. 3. FCC 3D (a) (x, y, z) 3D (b) 3D FCC (c) 4

. $\Xi_{m,i,j,k}$ m (i,j,k) $m \in \mathcal{S}_{active}$

$$\text{Hebbian} : \Xi^{t+1}_{m,i,j,k} = \max\left(\eta\, VT_m\, A^{g}_{i,j,k},\ \Xi^{t}_{m,i,j,k}\right) \tag{46}$$

$$: \Delta A^{g}_{i,j,k} \mathrel{+}= \frac{\rho}{|\mathcal{S}_{active}|} \sum_{m \in \mathcal{S}_{active}} VT_m\, \Xi_{m,i,j,k} \tag{47}$$

η Hebbian ρ $\mathcal{S}_{active}$
. HDC $N_\psi \times N_h$ × 2D MD-CAN
. $\mathbf{A}^{\mathrm{h}} \in \mathbb{R}^{N_\psi \times N_h}$ $N_\psi = 360$ $N_h = 5$
(u, h) $\theta_u = 2\pi u / N_\psi$ h
. HDC $\mathcal{K}^{\mathrm{h}}$

$$\Delta A^{\mathrm{h}}_{\psi,h} = \sum_{u,v} A^{\mathrm{h}}_{u,v}\, \mathcal{K}^{\mathrm{h}}_{\langle \psi - u \rangle_{N_\psi}, \langle h - v \rangle_{N_h}} \tag{48}$$

$$A^{\mathrm{h}} \leftarrow \frac{[A^{\mathrm{h}} + \Delta A^{\mathrm{h}} - \lambda_h \, \bar{A}^{\mathrm{h}}]_+}{\|A^{\mathrm{h}}\|_1} \tag{49}$$

r^f v_z^f

$$\Delta\psi = \lfloor \gamma_\psi r^f \Delta t \rfloor, \quad \Delta h = \lfloor \gamma_h v_z^f \Delta t \rfloor \tag{50}$$

γ_ψ γ_h
.

$$\hat{\mathbf{p}}_t = (\hat{x}_t, \hat{y}_t, \hat{z}_t) = \frac{\sum_{i,j,k} \mathbf{c}_{i,j,k} A^{\mathrm{g}}_{i,j,k,t}}{\sum_{i,j,k} A^{\mathrm{g}}_{i,j,k,t}}, \quad \mathbf{c}_{i,j,k} = (x_i, y_j, z_k) \tag{51}$$

$$\hat{\psi}_t = \operatorname{atan2}\left(\sum_{u,h} A^{\mathrm{h}}_{u,h,t} \sin\theta_u, \sum_{u,h} A^{\mathrm{h}}_{u,h,t} \cos\theta_u \right), \quad \theta_u = 2\pi u / N_\psi \tag{52}$$

atan2

E.

. EKF 10 $\log s$ KITTI 07 294/1101 VO ——
—
. t $\hat{z}_t$ Depth-Anything-V2 8

$$\mathbf{f}_t = \big[\log \hat{s}_t^{\mathrm{dav2}},\ \log d_t^{\mathrm{med}},\ \log(d_t^{\mathrm{std}}+\epsilon),\ \log d_t^{\mathrm{p10}},\ \log d_t^{\mathrm{p90}},\ \log(\hat{s}_t^{\mathrm{inl}}+1),\ f_x/W \big]^\top, \tag{53}$$

$\hat{s}_t^{\mathrm{dav2}}$ VO $d^{\mathrm{med}}, d^{\mathrm{p10}}, d^{\mathrm{p90}}, d^{\mathrm{std}}$ $\hat{s}_t^{\mathrm{inl}}$ —
f_x/W W=50 g_θ MLP 8→64→64→1 ReLU ∼4.8k 2 < 15 —— −0.12 m——
S.3

$$\hat{\ell}_t = g_\theta(\mathbf{f}_t) \ \approx\ \log s_t^*, \tag{54}$$

$$\ell_t^* = \log \frac{\|\mathbf{p}_t^* - \mathbf{p}_{t-1}^*\|}{\|\mathbf{p}_t - \mathbf{p}_{t-1}\|}, \tag{55}$$

$\mathbf{p}^*$ $\mathbf{p}$ VO
. Town05/03/07 Adam 10^{-3} 10^{-3} MSE 10% Town10HD
—
. $\hat{s}_t = \exp(g_\theta(\mathbf{f}_t))$ EKF (i) $\log s$ (ii) (iii) 10 ——

. —— —— sim-to-real KITTI 07 ≈ 1.8 $s \approx 0.3$ EKF S.1 —— ∼4.8k CPU 31-FPS

F.

3D -
. $\mathcal{E}_i$ $(VT_i,\ \boldsymbol{\phi}_i \in \mathbb{R}^{4096})$ $\mathbf{P}_i^{gc} = \mathbf{A}_i^{\mathrm{g}} \in \mathbb{R}^{N_x \times N_y \times N_z}$ $\mathbf{P}_i^{hdc} = \mathbf{A}_i^{\mathrm{h}} \in \mathbb{R}^{N_\psi \times N_h}$ (x_i, y_i, z_i, ψ_i) t_i
.

$$\begin{aligned} & d_{cos}(\boldsymbol{\phi}_t, \boldsymbol{\phi}_j) > \tau_{VT} \qquad j \in \mathcal{E} \\ & \|(\hat{x}_t, \hat{y}_t, \hat{z}_t) - (x_j, y_j, z_j)\| > d_{thresh} \end{aligned} \tag{56}$$

$\tau_{VT} = 0.08$ d_{thresh} 0.5 m

3D

$$\mathbf{T}_{i \to i+1} = (\Delta x, \Delta y, \Delta z, \Delta\psi) = (x_{i+1} - x_i, y_{i+1} - y_i, z_{i+1} - z_i, \psi_{i+1} - \psi_i) \tag{57}$$

$\mathcal{G} = (\mathcal{V}, \mathcal{E})$
.

$$\begin{aligned} & j : d_{cos}(\boldsymbol{\phi}_t, \boldsymbol{\phi}_j) < \tau_{loop} \\ & |t - t_j| > T_{min} \end{aligned} \tag{58}$$

$\tau_{loop} = 0.05$ $T_{min} = 50$ 10 Hz 5

$$\mathbf{T}_{loop} = (\hat{x}_t - x_j, \hat{y}_t - y_j, \hat{z}_t - z_j, \hat{\psi}_t - \psi_j) \tag{59}$$

.
1 . j t $\mathcal{P} = \{j, j+1, \ldots, t-1, t\}$
2 .

$$\mathbf{e}_{loop} = \sum_{k \in \mathcal{P}} \mathbf{T}_{k \to k+1} - \mathbf{T}_{loop} \tag{60}$$

$\mathbf{e}_{loop}$ $\mathcal{P}$
3 .

$$\mathbf{T}_{k \to k+1} \leftarrow \mathbf{T}_{k \to k+1} - \frac{\mathbf{e}_{loop}}{|\mathcal{P}|} \tag{61}$$

4 .

$$(x_{k+1}, y_{k+1}, z_{k+1}, \psi_{k+1}) = (x_k, y_k, z_k, \psi_k) + \mathbf{T}_{k \to k+1} \tag{62}$$

5 . $(\hat{x}, \hat{y}, \hat{z}, \hat{\psi})_{corr}$ “ ”
. DINOv2-S $\mathbf{e}_t$ $j^* = \arg\max_j \cos(\mathbf{e}_t, \mathbf{e}_j)$ EKF $\mathrm{arc}_{\min}$ 50 m ——
. —

$$\mathrm{accept}(t, j^*) \iff \cos(\mathbf{e}_t, \mathbf{e}_{j^*}) \geq \tau \wedge \mathrm{streak}(t, j^*) \geq 15 \wedge |\Delta\psi_{tj^*}| \leq 60^\circ \wedge \| \sum_i \mathbf{c}_i^\top \| \leq 35 \text{ m} \tag{63}$$

$\tau = 0.94$ 0.907–0.935 ≥ 0.949 ≥ 15 ±10
— 60° KITTI 09
. $\mathbf{c}_{tj^*}$ $\beta = 1.0$ link
Sim(3)
. f_i $\mathbf{c}_i$ $\mathbf{C}_{\mathrm{cum}}(f)$ - $\|\mathbf{c}_i\| \leq C_{\max}$ W_i

$$F_i(f) = \begin{cases} \frac{1}{2}\big(1 - \cos(\pi(f - f_i)/W_i)\big), & f_i \leq f < f_i + W_i \\ 1, & f \geq f_i + W_i \end{cases} \tag{64}$$

—

$$\hat{\mathbf{p}}(f) = \mathbf{p}_{\exp}(f) + \Big(\sum_i F_i(f)\, \mathbf{c}_i \ - \sum_{i:\, f_i \leq f} \mathbf{c}_i \Big), \tag{65}$$

$\mathbf{c}_i$ W_i $\mathbf{C}_{\mathrm{cum}}$

- . $W_i = \mathrm{clip}(W_0 + W_\perp \|\mathbf{c}_i\|,\ W_0,\ W_{\max})$
- . $W_i \leftarrow \min(W_i, f_{\mathrm{end}} - f_i + 1)$
- . $\|\mathbf{c}_i\| > C_{\max}$

Town01 RMSE +0.64% EKF
EKF NLM_USE_EKF_ODO EKF EKF
Sim(3) .
t_{now} $\mathbf{A}$ win $\mathbf{C}$ t_{now} win Procrustes

$$\mathbf{A} \approx s\,(\mathbf{C}\mathbf{R}) + \mathbf{t}, \qquad s \in [0.05,\ 20], \tag{66}$$

win = 250 t_{now}

$$\hat{\mathbf{p}}(f) = s\,(\mathbf{p}(f)\mathbf{R}) + \mathbf{t}, \qquad t_{\mathrm{now}} < f < t_{\mathrm{next}}, \tag{67}$$

843.8 m Sim(3) EKF 9× 44.24 m 398.05 m Sim(3) SE(3) 193.23 m 4.4×——

IV.

CARLA Town05/03/07/10HD KITTI 07 NeuroLocMap - EKF

A.

. Intel Core i5-12450H CPU 12 8 16 GB NVIDIA GeForce RTX 4050 GPU GPU HART Transformer CPU LSTM S

. NeuroLocMap (1) EKF —— 10 $(x, y, z, v_x, v_y, v_z, \phi, \theta, \psi, \log s)^\top$ 10 $\log s$ VO 0 IMU (i) VO × IMU $\times\Delta t$ (ii) / (iii) 30 × VO × VO (iv) $|a_{\text{lat}}|/|\omega|$ VO $\log s$ Mahalanobis 0.99 R $[R_{\min}, R_{\max}]$ VO Joseph + MSCKF/VINS-Mono [15], [16] - (2) VO —— ORB VO + recoverPose IMU EKF+ 10 EKF

. Python CARLA EKF fusion_pose.txt VO visual_odometry.txt ground_truth.txt - NeuroLocMap CARLA EKF

. II (1) CARLA Town05/03/07/10HD Town05 486 m Town03 474 m Town07 618 m Town10HD 1451 m KITTI 07 [35] 692 m 1101

. SLAM ATE RMSE [36] RPE NeuroLocMap (i) Sim(3) Umeyama/Procrustes —— (ii) SE(3) Kabsch —— SE(3) NeuroLocMap EKF Sim(3) 0.79–1.92 Sim(3) " "

B.

. III NeuroLocMap 10 EKF - Sim(3) +16.5% KITTI 07 +59.3% Town05 SE(3) +1.5% KITTI 07 +33.9% Town05 —— EKF SE(3) 13.1–32.7% Sim(3) 22.4–54.8

. CARLA 452–618 m NeuroLocMap EKF Sim(3) 22.4% Town10HD 59.3% Town05 SE(3) 13.1% 33.9% VO 42.6–68.8% EKF→EKF+ 0–2.0 m Town05/03/07 7–8 link Town10HD 1 link 90–481 Town05: 175 Town03: 90 Town07: 269 Town10HD: 481 67–481 692 m KITTI 07 NeuroLocMap EKF S.1 Sim(3) 16.5% 18.45 m 22.09 m SE(3) 1.5% 29.99 m 30.45 m EKF 33.7% 27.84 m VO - ——

. 4 Town05 486 m Town10HD 452 m Town05 NeuroLocMap 18.07 m RMSE—— EKF 44.44 m 59.3% 24.2% 4.6% 62.6%—— 175 /175 7 link Town10HD 14.44 m EKF 18.61 m +22.4% VO 35.86 m +59.7% 481 1 link EKF+ 14.44 m EKF+ III

. CARLA 90–481 Town05: 175 Town03: 90 Town07: 269 Town10HD: 481 67–481 1–8 PR link KITTI 07 692 m 964 929 ∼1.3 / 1.5 m 12 m IMU

. IV NeuroLocMap 31 FPS 32.1 ms 3D 58% 15% 7% $\mathcal{O}(N)$ 31-FPS <1 ms CPU

. (1) $\mathcal{O}(HW)$ $\mathcal{O}(N_f)$ $H \times W = 640 \times 480$ $N_f \approx 300$ (2) 3D $\mathcal{O}(N_x N_y N_z)$ $30 \times 30 \times 10 = 9000$ (3) $\mathcal{O}(M \cdot d)$ M 67–929 d 4096 (4) $\mathcal{O}(K)$ K 90–964 $\mathcal{O}(HW + N_x N_y N_z + Md)$ GPU $\mathcal{O}(N^2)$ $\mathcal{O}(N)$ CPU

. SLAM GPU Intel Core i5-12450H CPU 8 2.0–4.4 GHz 16 GB 2 GB 3D 15 MB 929 × 4096 × 4 CPU 45–60% NVIDIA Jetson Nano 4 —— GPU SLAM DROID-SLAM 8 GB TartanVO 6 GB 15–20 W CPU GPU 50–80 W

C.

. 5(a) Town05 486 m Sim(3) 18.07 m EKF+ 20.10 m −0.55 =0.6 EKF 41.14 m 31 FPS- 44.44 m VO 48.26 m EKF 26.4 m 24.3 m −2.03 m—— IV-E Town03 15.31 / EKF+ 16.07 / 22.91 / EKF 23.36 m Town10HD ≡ EKF+ 14.44 m

. ORB Town05 ORB 44.72 m—— EKF 44.44 m —— 41.14 m Sim(3) 0.99– 5(b) 67–929 RatSLAM ∼5 [37] KITTI 07 692 m 929 ∼1.3 / CARLA 67–481 Town05: 175 Town10HD: 481 Town07: 263 Town03: 67

D.

KITTI 07 EKF/VO 12 km/h 6 m 20 20 III

. (a) NeuroLocMap Town07 Sim(3) EKF+ (b) Sim(3) EKF +16.5% +59.3% (c) VO IMU +33.7% +68.8% [34] Town05 Town07 618 m Town10HD 452 m Town10HD Sim(3) 1451 1.92 (2)Town05/03/07 7–8 link Town10HD link NeuroLocMap

- —— VO Town10HD EKF 0.65 35% 18.61 m Sim(3) / 57.38 m SE(3) 14.44 m / 49.88 m SE(3) Sim(3) otherwise
- —— Town05 7 link −2.03 m Town03 8 link −0.76 m Town10HD link —— EKF+
- ——KITTI 07 +16.5% Sim(3) / +1.5% SE(3) S.1

E. EKF EKF+

Town07 Town07 28.75 m Sim(3) / 31.51 m SE(3) EKF+ 27.90/29.21 m +0.84 m $\tau = 0.94$ 15 60° Town07

KITTI 07 CARLA KITTI $s \approx 0.3$ ≈ 1.8 S.1 KITTI VIO EKF 17.8 m 23.1 m Sim(3) KITTI sim-to-real —— KITTI Town10HD link EKF+ 14.44 m . NeuroLocMap —— 452–618 m CARLA 692 m RMSE RPE / VO 48.26 m

F. -

7 KITTI 07 1101 692 m - (a) IMU (b) ——NeuroLocMap 18.45 m RMSE EKF 22.09 m VO 27.84 m Sim(3) —— 964 929 link 11 / EKF VIO/SLAM . NeuroLocMap EKF/VO V KITTI —— RTE 100–800 m % III / —— - SLAM KITTI 07 2.1 ms 469 Hz RTE 1.2% NeuroLocMap +IMU S.1 KITTI RTE 94.7/101.1% GT 1.9 ——7 Sim(3) —— RTE 72.1/75.2% 50 38.2/46.9% S.1 up-to-scale VIO / KITTI 07 EKF ATE 22.09 m 18.45 m −16.5% CARLA EKF ATE 59.3% III NeuroLocMap up-to-scale VIO

TABLE II
CARLA 12 km/h EKF 20 + 20 KITTI " link "

Town05	486 m	5000	7 link
Town03	474 m	5000	8 link
Town07	618 m	5000	7 link
Town10HD	452 m	5000	1 link
KITTI 07	692 m	1101	11 link

TABLE III
ATE RMSE m

	Sim(3)				SE(3)			
	NeuroLocMap	EKF+	EKF	VO	NeuroLocMap	EKF+	EKF	VO
Town05	18.07	20.10	44.44	48.26	32.67	33.26	49.40	53.86
Town03	15.31	16.07	23.36	49.10	17.31	17.47	23.53	49.15
Town07	28.75	27.90	40.64	50.07	31.51	29.21	42.85	50.13
Town10HD	14.44	14.44	18.61	35.86	49.88	49.89	57.38	58.04
KITTI 07	18.45	n/a	22.09	27.84	29.99	n/a	30.45	38.35

= NeuroLocMap EKF Sim(3) +16.5% +59.3% SE(3) +1.5% +33.9% VO Sim(3) +33.7% +68.8%
/ NeuroLocMap 2.9–11.7% Sim(3) 0.2–15.2% SE(3) EKF 1.6–13.3% 5.0–19.0% VO 9.9–24.2% 13.7–17.2%
n/a KITTI 07 S.1 KITTI EKF Town07 +0.84 m

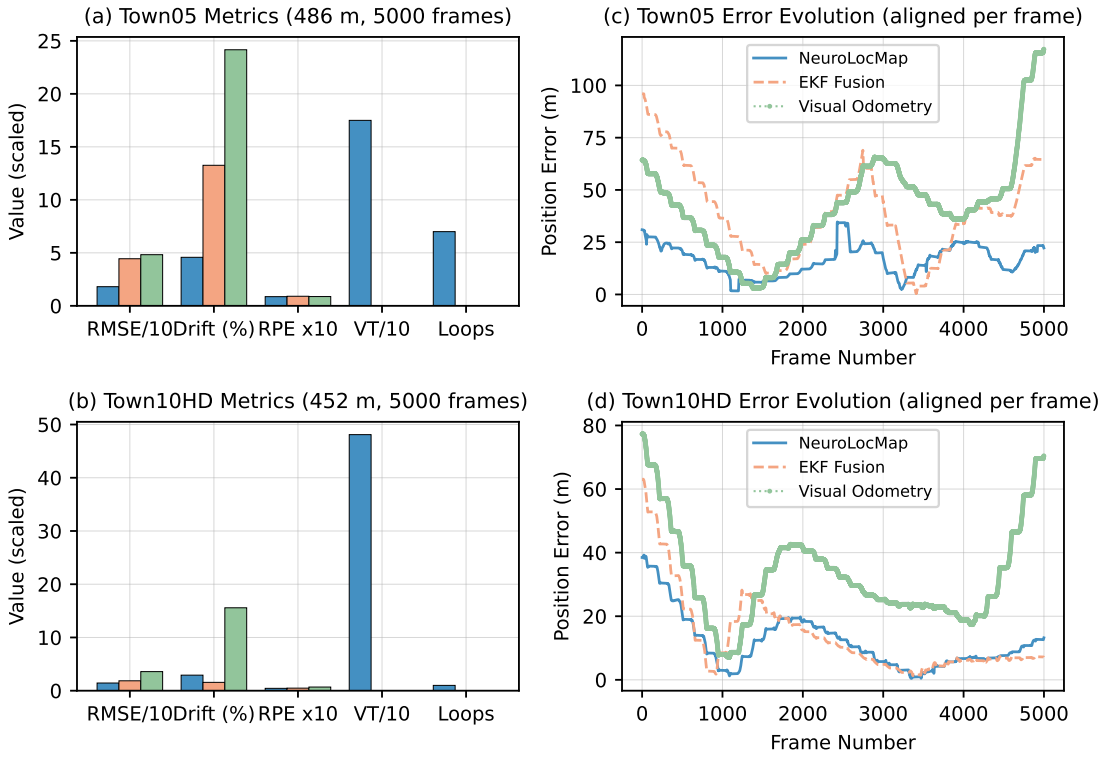


Fig. 4. Town05 486 m Town10HD 452 m RMSE RPE / Town05 NeuroLocMap 18.07 m EKF 44.44 m 59.3% VO 48.26 m 62.6% Town10HD 14.44 m EKF 18.61 m +22.4% EKF+ link

. SLAM DROID-SLAM [42] TartanVO [43] DeepVO [44] TartanVO 100 GPU

- DeepVO KITTI 07 10.2% NeuroLocMap 9.71% TartanVO CARLA 2–3% NeuroLocMap 2.9–11.7%
- DROID-SLAM 4–5 FPS NeuroLocMap ∼31 FPS
- NeuroLocMap HDC
- NeuroLocMap

. NeuroLocMap

- Sim-to-Real S.1
- 4 6
- KITTI 07 929 ∼1.3 /
- Town07

TABLE IV
CARLA

	(ms)	(Hz)
	2.1	469
	2.3	429
-	1.9	527
3D CAN	18.5	54
HDC	1.5	661
	4.7	213
	32.1	31.2

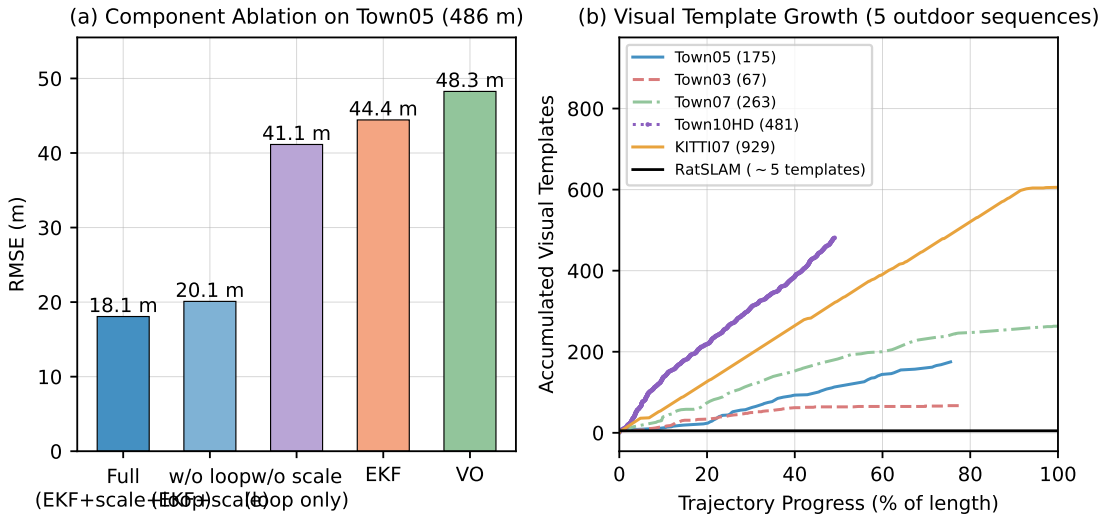


Fig. 5. (a) Town05 486 m Sim(3) EKF+ + 18.07 m w/o EKF+ 20.10 m w/o 41.14 m EKF 44.44 m VO 48.26 m (b) RatSLAM ∼5 [37]

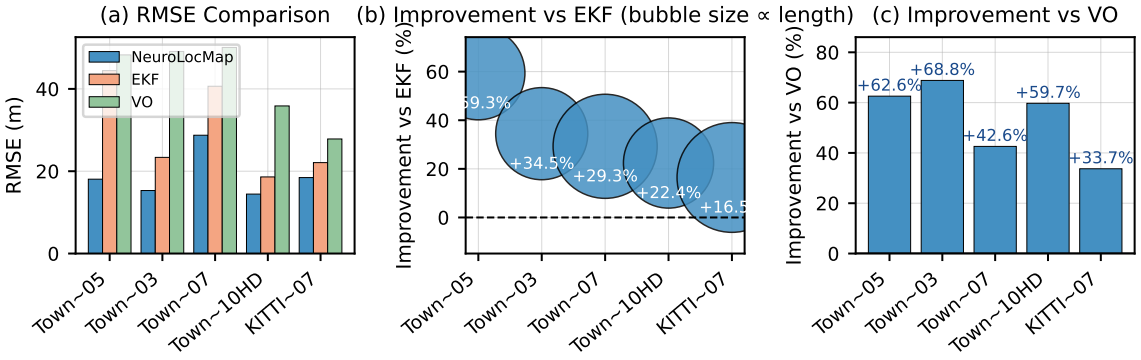


Fig. 6. (a) NeuroLocMap EKF+ EKF VO RMSE Sim(3) (b) EKF +16.5% +59.3% (c) VO +33.7% +68.8%

- $\tau = 0.94$

S.1 KITTI 07

CARLA KITTI 07 $\log s \approx -1.1$ $s \approx 0.3$ VO $s \approx 1.8$ —— CARLA 0.30 replay EKF —— VIO EKF 17.75 m 23.13 m Sim(3) VO 5.5 0.5 $s \approx 5.65$ VO EKF ≈ 5.54 ≈ 6.78—— KITTI EKF KITTI

S.2 KITTI 07

KITTI 07 10 EKF IMU — kitti_prepare_fusion.py ORB EKF OXTS IMU

KITTI 07	Sim(3)	SE(3)
EKF	22.09 m	30.45 m
NeuroLocMap	18.45 m	29.99 m
EKF VIO	17.75 m	44.01 m
NeuroLocMap	23.12 m	40.99 m

Sim(3) VIO 17.75 m 23.12 m Sim(3)

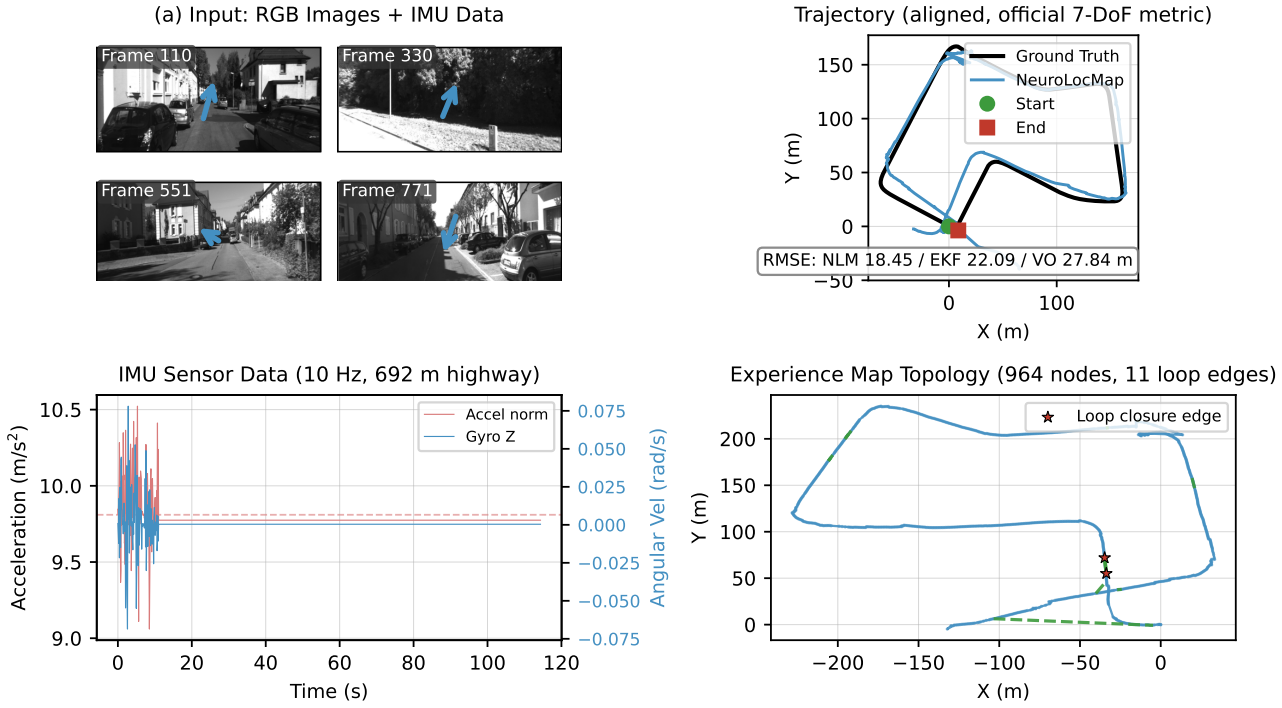


Fig. 7. KITTI 07 692 m 1101 - (a) IMU (b) vs NeuroLocMap RMSE 18.45 m EKF 22.09 m Sim(3) 964 11 /

TABLE V
KITTI RTE % ATE 07 7 Sim(3) RMSE m

		RTE (%)	ATE (m)
SOFT-SLAM [38]		0.65	—
VINS-Fusion [39]	+IMU	1.09	—
ORB-SLAM2 [40]		1.15	—
MonoDepth2 [41]		12.59	—
NeuroLocMap EKF	+IMU	94.7 / 72.1	22.09
NeuroLocMap	+IMU	101.1 / 75.2	18.45

RTE KITTI MonoDepth2 NeuroLocMap RTE 07 / 7 Sim(3) ATE Sim(3) RMSE

—— 1.8 —— SE(3) —— —— 40.99 44.01 m −6.8% KITTI

S.3 KITTI 09

KITTI 09 1591 1705.1 m EKF Sim(3) 42.37 42.34 m SE(3) 54.63 54.77 m —— GT 3.8 m 0.988 0.994 15 2 what-if −0.12 m Sim(3) —— GT 90/134 m 964 929 —— CARLA —— KITTI 09 0% 100% KITTI 07

S.4

CARLA 12 km/h 20 20 KITTI 07 KITTI 09 kitti09_assemble_raw.py EKF kitti_prepare_fusion.py KITTI 07 S.2 VIO se3_metrics.py

V.

NeuroLocMap—— 4 - 3D - / Town 05 / 03 / 07 / 10HD KITTI 07 NeuroLocMap 10 EKF - —— Sim(3) +16.5–59.3% SE(3) 33.9%—— VO +33.7–68.8% 31 FPS EKF 9 44. 398.05 m Sim(3) 5 RatSLAM 929 185 —— —— sim-to-real 6 to-real

2024JJ6190 22B0646 24XJJCY 2023SZJ19

NeuroLocMap https://github.com/OpenHUTB/carla-pedestrians

Haidong Wang Caixia Ning

References

[1] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard, “Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age,” IEEE Transactions on Robotics, vol. 32, no. 6, pp. 1309–1332, 2016.

[2] M. R. U. Saputra, A. Markham, and N. Trigoni, “Visual slam and structure from motion in dynamic environments: A survey,” ACM Computing Surveys, vol. 51, no. 2, pp. 1–36, 2018.

[3] K. J. Jeffery, A. Jovalekic, M. Verriotis, and R. Hayman, "Navigating in a three-dimensional world," Behavioral and Brain Sciences, vol. 36, no. 5, pp. 523–543, 2013.

[4] T. Hafting, M. Fyhn, S. Molden, M.-B. Moser, and E. I. Moser, "Microstructure of a spatial map in the entorhinal cortex," Nature, vol. 436, no. 7052, pp. 801–806, 2005.

[5] J. S. Taube, R. U. Muller, and J. B. Ranck, "Head-direction cells recorded from the postsubiculum in freely moving rats. i. description and quantitative analysis," Journal of Neuroscience, vol. 10, no. 2, pp. 420–435, 1990.

[6] A. Finkelstein, D. Derdikman, A. Rubin, J. N. Foerster, L. Las, and N. Ulanovsky, "Three-dimensional head-direction coding in the bat brain," Nature, vol. 517, no. 7533, pp. 159–164, 2016.

[7] T. Solstad, C. N. Boccara, E. Kropff, M.-B. Moser, and E. I. Moser, "Representation of geometric borders in the entorhinal cortex," Science, vol. 322, no. 5909, pp. 1865–1868, 2008.

[8] M. J. Milford, G. F. Wyeth, and D. Prasser, "Ratslam: A hippocampal model for simultaneous localization and mapping," IEEE International Conference on Robotics and Automation, pp. 403–408, 2004.

[9] M. Milford and G. Wyeth, "Persistent navigation and mapping using a biologically inspired slam system," The International Journal of Robotics Research, vol. 29, no. 9, pp. 1131–1153, 2010.

[10] R. Mur-Artal, J. M. M. Montiel, and J. D. Tardos, "Orb-slam: A versatile and accurate monocular slam system," IEEE Transactions on Robotics, vol. 31, no. 5, pp. 1147–1163, 2015.

[11] J. Engel, T. Schöps, and D. Cremers, "Lsd-slam: Large-scale direct monocular slam," in European Conference on Computer Vision. Springer, 2014, pp. 834–849.

[12] J. Engel, V. Koltun, and D. Cremers, "Direct sparse odometry," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 40, no. 3, pp. 611–625, 2018.

[13] M. Cummins and P. Newman, "FAB-MAP: Probabilistic localization and mapping in the space of appearance," The International Journal of Robotics Research, vol. 27, no. 6, pp. 647–665, 2008.

[14] R. Arandjelovic, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 5297–5307.

[15] A. I. Mourikis and S. I. Roumeliotis, "A multi-state constraint kalman filter for vision-aided inertial navigation," in Proceedings 2007 IEEE International Conference on Robotics and Automation. IEEE, 2007, pp. 3565–3572.

[16] T. Qin, P. Li, and S. Shen, "VINS-Mono: A robust and versatile monocular visual-inertial state estimator," IEEE Transactions on Robotics, vol. 34, no. 4, pp. 1004–1020, 2018.

[17] R. Mahony, T. Hamel, and J. Pflimlin, "Nonlinear complementary filters on the special orthogonal group," IEEE Transactions on Automatic Control, vol. 53, pp. 1203–1218, 2008.

[18] S. O. H. Madgwick, A. J. L. Harrison, and R. Vaidyanathan, "Estimation of imu and marg orientation using a gradient descent algorithm," IEEE International Conference on Rehabilitation Robotics, pp. 1–7, 2011.

[19] D. E. Angelaki and K. E. Cullen, "Vestibular system: the many facets of a multimodal sense," Annual Review of Neuroscience, vol. 27, pp. 393–418, 2004.

[20] K. E. Cullen, "The vestibular system: multimodal integration and encoding of self-motion for motor control," Trends in Neurosciences, vol. 35, no. 3, pp. 185–196, 2012.

[21] J. O'Keefe and J. Dostrovsky, "The hippocampus as a spatial map: preliminary evidence from unit activity in the freely-moving rat," Brain Research, vol. 34, no. 1, pp. 171–175, 1971.

[22] T. Hafting, M. Fyhn, S. Molden, M.-B. Moser, and E. I. Moser, "Microstructure of a spatial map in the entorhinal cortex," Nature, vol. 436, no. 7052, pp. 801–806, 2005.

[23] J. S. Taube, R. U. Muller, and J. B. Ranck, "Head-direction cells recorded from the postsubiculum in freely moving rats. i. description and quantitative analysis," Journal of Neuroscience, vol. 10, no. 2, pp. 420–435, 1990.

[24] E. I. Moser, M.-B. Moser, and B. L. McNaughton, "Spatial representation in the hippocampal formation: a history," Nature neuroscience, vol. 20, no. 11, pp. 1448–1464, 2017.

[25] A. Finkelstein, D. Derdikman, A. Rubin, J. N. Foerster, L. Las, and N. Ulanovsky, "Three-dimensional head-direction coding in the bat brain," Nature, vol. 517, no. 7533, pp. 159–164, 2015.

[26] M. J. Milford and G. F. Wyeth, "Mapping a suburb with a single camera using a biologically inspired slam system," IEEE Transactions on Robotics, vol. 24, no. 5, pp. 1038–1053, 2008.

[27] J. Steckel and H. Peremans, "Batslam: Simultaneous localization and mapping using biomimetic sonar," PLoS One, vol. 8, no. 1, p. e54076, 2013.

[28] S. Lowry, N. Sünderhauf, P. Newman, J. J. Leonard, D. Cox, P. Corke, and M. J. Milford, "Visual place recognition: A survey," IEEE Transactions on Robotics, vol. 32, no. 1, pp. 1–19, 2016.

[29] M. A. Goodale and A. D. Milner, "Separate visual pathways for perception and action," Trends in Neurosciences, vol. 15, no. 1, pp. 20–25, 1992.

[30] M. Mishkin, L. G. Ungerleider, and K. A. Macko, "Object vision and spatial vision: two cortical pathways," Trends in Neurosciences, vol. 6, pp. 414–417, 1983.

[31] J. J. DiCarlo, D. Zoccolan, and N. C. Rust, "How does the brain solve visual object recognition?" Neuron, vol. 73, no. 3, pp. 415–434, 2012.

[32] C. J. Duffy and R. H. Wurtz, "Sensitivity of mst neurons to optic flow stimuli. ii. mechanisms of response selectivity revealed by small-field stimuli," Journal of Neurophysiology, vol. 65, no. 6, pp. 1346–1359, 1998.

[33] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 5998–6008.

[34] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "Carla: An open urban driving simulator," in Conference on Robot Learning. PMLR, 2017, pp. 1–16.

[35] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? the kitti vision benchmark suite," in 2012 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2012, pp. 3354–3361.

[36] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, "A benchmark for the evaluation of rgb-d slam systems," in 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2012, pp. 573–580.

[37] M. J. Milford and G. F. Wyeth, "SeqSLAM: Visual route-based navigation for sunny summer days and stormy winter nights," in 2012 IEEE International Conference on Robotics and Automation. IEEE, 2012, pp. 1643–1649.

[38] I. Cvišić, J. Česić, I. Marković, and I. Petrović, "Soft-slam: Computationally efficient stereo visual slam for autonomous uavs," Journal of Field Robotics, 2017.

[39] T. Qin, J. Pan, S. Cao, and S. Shen, "Vins-fusion: A robust and versatile visual-inertial-state estimator with automatic configuration," arXiv preprint arXiv:1808.00903, 2018.

[40] R. Mur-Artal and J. D. Tardós, "ORB-SLAM2: An open-source SLAM system for monocular, stereo, and RGB-D cameras," IEEE transactions on robotics, vol. 33, no. 5, pp. 1255–1262, 2017.

[41] C. Godard, C. K. MacKenzie, and G. J. Davison, "Digging deeper into no-stereo depth estimation," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021.

[42] Z. Teed and J. Deng, "Droid-slam: Deep visual slam for monocular, stereo, and rgb-d cameras," in Advances in Neural Information Processing Systems (NeurIPS), vol. 34, 2021, pp. 16 558–16 569.

[43] W. Wang, Y. Hu, and S. Scherer, "Tartanvo: A generalizable learning-based vo," in Conference on Robot Learning (CoRL), 2021, pp. 1761–1772.

[44] S. Wang, R. Clark, H. Wen, and N. Trigoni, "Deepvo: Towards end-to-end visual odometry with deep recurrent convolutional neural networks," in 2017 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2017, pp. 2043–2050.